



N109AV (S/N: W03021891423) — Modem Stuck in Low-Power Mode Diagnostic Report

Immfly Ground Operations / IT — Session: August 31–September 1, 2026

Revision note: This report supersedes two earlier findings for this unit. The first identified the SIM's PIN2/Fixed Dialing Numbers (FDN) lock as the root cause; this was disproven by cross-checking against two known-healthy units (N064AV and N140AV), both of which carry the identical PIN2/FDN lock while functioning normally. The second, produced shortly afterward, identified the modem as physically absent from the USB bus; this was also superseded when, following an unrelated physical repair on the unit's Teensy front-display board and a subsequent reboot, the modem reappeared and enumerated normally on the USB bus. Further testing after that point revealed the actual, current issue described below.

Field	Value
Machine	N109AV (S/N: W03021891423)
Affected component	Sierra Wireless EM7565 modem — stuck in low-power operating mode
Ruled out	SIM PIN2/FDN lock (normal fleet-wide condition); modem absent from USB bus (resolved itself after an unrelated reboot)
Root cause	Modem enumerates normally on USB (1199:9091) but remains in 'low-power' operating mode and cannot transition to 'online'/enabled — every enable attempt via ModemManager times out with a QMI transaction timeout
Recovery attempted	ModemManager restart, full machine power cycle, and a direct qmicli operating-mode change — none succeeded; the direct qmicli attempt failed with the same 'CID allocation failed / endpoint hangup' error previously seen right before another unit (N060AV) became stuck in QDL recovery mode
Risk noted	Repeating or escalating low-level qmicli operating-mode commands carries a demonstrated risk of pushing the modem into QDL (download/recovery) mode, a more severe and, in the N060AV case, unrecoverable state — further blind attempts were deliberately avoided
Recommended action	Escalate for physical inspection / modem module evaluation rather than continuing low-level software recovery attempts on-site

Narrative Summary

N109AV's modem troubleshooting went through three distinct phases over this session. Initially, the modem failed to enable with an 'Invalid transition' error; this was traced in part to 26+ accumulated zombie pppd processes (cleared), and a SIM PIN2/FDN lock was suspected as the remaining cause. That suspicion was disproven by comparing against two confirmed-healthy units showing the same lock state.

Re-investigating N109AV directly then revealed the modem was not enumerating on the USB bus at all — a more serious finding, matching the failure signature previously documented on N120AV. However, an unrelated physical repair was performed on the unit's Teensy front-display board around this time, and following the machine reboot that came with that repair, the modem reappeared and enumerated normally on the USB bus

(1199:9091, with all three expected serial ports and the QMI/net interfaces present). This indicates the earlier total absence from the bus was a transient condition — possibly related to the physical work being done on the unit at the time — rather than a permanent hardware fault.

With the modem physically present again, a new and distinct issue emerged: `mmcli -m 0 --enable` consistently failed with a **QMI transaction timeout** error, and the modem's own reported operating mode (`qmicli --dms-get-operating-mode`) remained stuck at '**low-power**' rather than progressing to 'online'. This is a different failure mode from the earlier bus-absence issue: the modem is physically present and responds to some queries, but cannot complete the transition needed to become operational.

A restart of the ModemManager service and a full physical power cycle of the machine were both attempted; the timeout persisted identically after each. A direct, lower-level attempt to force the operating mode via `qmicli --dms-set-operating-mode=online` was then made, bypassing ModemManager entirely. This command failed with '**CID allocation failed in the CTL client: endpoint hangup**' — notably, this is the same error signature observed immediately before another unit in this fleet (N060AV) became stuck in QDL (Qualcomm download/recovery) mode during a firmware operation, a state from which recovery was ultimately unsuccessful in that case.

The modem's USB enumeration was immediately re-checked after this attempt and confirmed to still be in normal operating mode (1199:9091, not the QDL 1199:9090 signature). Given this demonstrated risk, further low-level operating-mode commands were deliberately not attempted, to avoid repeating the sequence of events that led to N060AV's unrecoverable QDL state.

Troubleshooting Steps Performed

1. Initial 'Invalid transition' error and zombie process cleanup

26+ accumulated pppd processes were found competing for the modem's serial port and were cleared; the connectivity cron job was paused. The enable error persisted, ruling this out as the sole cause.

2. SIM PIN2/FDN lock investigated and ruled out

lock: sim-pin2 / enabled locks: fixed-dialing was observed and initially suspected as the root cause, then disproven by confirming the identical lock state on two known-healthy units (N064AV, N140AV).

3. Modem found absent from USB bus

/dev/modemPPP and /dev/modemAT did not exist; dmesg showed no Sierra Wireless device enumeration at all; lsusb confirmed no device with vendor ID 1199 present.

4. Unrelated physical repair and reboot

A physical repair was performed on the unit's Teensy front-display board, followed by a machine reboot. After this reboot, the modem reappeared and enumerated normally on the USB bus (1199:9091, all expected serial ports present).

5. Enable attempt on the now-present modem

mmcli -m 0 --enable failed with 'GDBus.Error:org.freedesktop.libqmi.Error.Core.Timeout: QMI operation failed: Transaction timed out'. Modem remained in state: disabled, power state: low.

6. ModemManager restart

systemctl restart ModemManager followed by another enable attempt produced the identical timeout error.

7. Full machine reboot

A complete power-off/power-on cycle was performed. Post-reboot, the modem was still detected (1199:9091) and mmcli -L found it, but the enable attempt again failed with the identical QMI transaction timeout.

8. Operating mode check

qmicli -d /dev/cdc-wdm0 --dms-get-operating-mode confirmed the modem was in 'low-power' mode, consistent with the enable failures.

9. Direct qmicli operating-mode change attempt

qmicli -d /dev/cdc-wdm0 --dms-set-operating-mode=online was attempted to bypass ModemManager. This failed with 'CID allocation failed in the CTL client: endpoint hangup' — the same error signature seen immediately before N060AV entered QDL mode during a separate firmware-recovery session.

10. Immediate safety check and decision to stop

lsusb was immediately re-checked and confirmed the modem remained in normal mode (1199:9091, not the QDL signature 1199:9090). Given the demonstrated risk of this class of command, no further low-level operating-mode attempts were made.

Conclusion and Recommendation

N109AV's modem is physically present and enumerates normally on the USB bus, but is stuck in low-power operating mode and cannot be transitioned to an operational state through either ModemManager or a direct qmicli command — every attempt results in a QMI transaction timeout, even after a full machine power cycle. A direct attempt to force the operating mode failed with an error signature that, on another unit in this fleet, preceded an unrecoverable QDL (download/recovery) mode failure. This risk was recognized in time and the

modem's operational mode was confirmed unaffected, but it means this specific recovery path should not be repeated on-site without a safer, more controlled procedure.

Recommended action: Escalate N109AV for physical inspection and modem module evaluation rather than continuing low-level QMI recovery attempts on-site, given the demonstrated risk of pushing the modem into an unrecoverable QDL state (as previously seen on N060AV). If a bench environment with safer recovery tooling is available, a controlled firmware/mode-reset attempt could be considered there; otherwise, physical modem replacement is the more reliable path back to service for this unit.